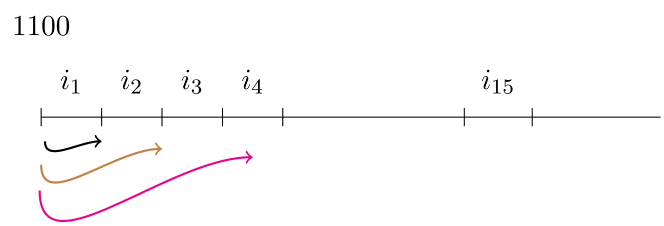
Problem 4.23

What is $A(15)$, if $A(0) = \$1{,}100$, and the rate of simple interest of the period n is $i_n = 0.01n$?

Solution

The amount $A(15)$ is the initial capital plus the interest earned in each period until reaching period 15.



Then,

$$A(15) = A(0) + \sum_{n=1}^{15} \big(A(0) \times (0.01n)\big).$$

The following table shows each term that appears in the sum.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$11n$	11	22	33	44	55	66	77	88	99	110	121	132	143	154	165

So,

$$A(15) = 1100 + 1320.$$

Therefore,

$$A(15) = 2420.$$